Grade B

Full Custom DAC Design Coursework

Yubo Zhi (yz4116)

***Abstract*—This coursework is about design a Digital to Analogue Converter (DAC) using full custom CMOS process. The design specifications was successfully achieved by using capacitor array approach. A compact layout was also designed using various techniques.**

I. Introduction

DIGITAL to Analogue Converter is an essential interface component heavily used in modern technology. It generates an analogue voltage signal based on its digital inputs. In this coursework [1], a DAC with 8-bit precision and serial data input was designed using full custom CMOS process, so that it can potentially be used as a component in integrated circuits.

December 16, 2016

II. System Overview

There are 3 possible topologies for this coursework [1], R-2R Ladder based, capacitor array based or current steering based [2].

It would be quit difficult to match resistors for R-2R Ladder topology. Moreover, parasitic resistances of MOSFETs may significantly affect the accuracy.

For capacitor array and current steering topology, the size of capacitors or the width of MOSFETs need to be doubled for each bit. For the specified 8-bit resolution, that would require about 512 times the layout area of a unit component. The current steering topology will also be affected by MOSFET parasitic resistances, whereas the capacitor array topology will only be affected by parasitic capacitances, which can be insignificant compare to the unit capacitor for an 8-bit DAC.

Further more, a split capacitor can be used in the capacitor array topology. By splitting the capacitors to 2 groups of 4 bits, only about 32 times the layout area of a unit capacitor would be needed. Common centroid and dummy devices layout techniques can also be used to improve component matching. By not having a continuous current flow comparing to the other 2 topologies, it would have a lower static power consumption. Therefore the capacitor array approach with a split capacitor was used in this coursework.

Fig. 1 shows the system block diagram of the proposed DAC design.

Data clock is generated inside the DAC module by a 10MHz ring oscillator.

Serial data input and enable signal are captured by 2 D-type flip-flops (DFF) at raising edge of the clock. These DFFs are used to avoid metastability issues, when inputs are changing at the same time the clock signal is rising. They will also hold the input for the rest of the clock cycle.

The serial data input then enters an 8-bit shift register. It will register each bit and output as an 8-bit parallel interface

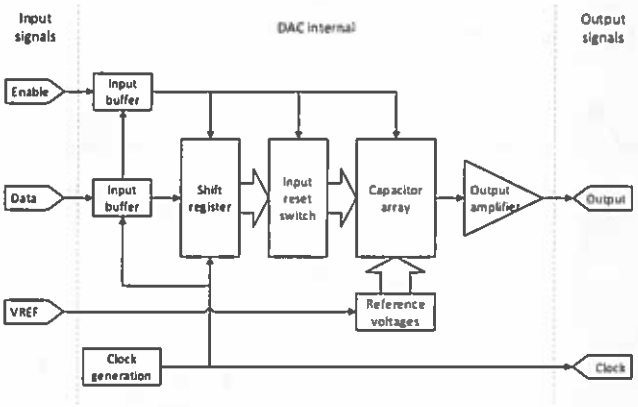


Fig. 1. System block diagram

for the capacitor array. The clock to shift register was gated by the enable signal, so it will hold the output when enable is not asserted.

The capacitor array consists of binary weighted capacitors. It works by charge redistribution that happens when different capacitors are hold at different voltage levels, with another terminal connected together. Therefore, it is essential to always release the existing charges on the capacitors before every conversion. Thus another input switch that connects the capacitor array inputs to ground during reset is needed.

According to specification, the output range should be from $0.4V$ up to $1.4V$. Therefore, voltage references are needed to biasing the capacitor array, so that it works at the specified voltage range. Overall gain and offset errors can also be compensated by adjusting the voltage references.

The output impedance of the capacitor array is very high, as there are only capacitors connected. Therefore, an output amplifier is required.

III. Circuit Implementation

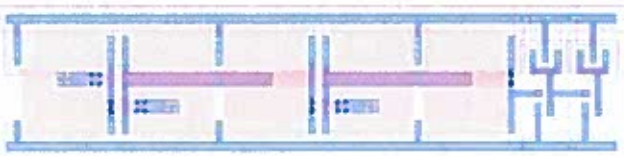
A. Clock generation

The 10MHz digital clock is generated by using a ring oscillator. Fig. 2 shows the schematic and layout design of the ring oscillator. Long channel MOSFETs are used to reduce oscillation frequency. 2 inverters are used to amplify and shape the output. Simulations are used to tune MOSFET parameters.

The spacing between N+ implant and the cell boundary is $0.2\mu m$, half the design rule for N+ implant spacing ($0.4\mu m$), same as others. It also has the same cell height of $8\mu m$ as other digital blocks, so it can be placed tightly with other digital blocks.



(a) Schematic



(b) Layout

Fig. 2. Ring oscillator design

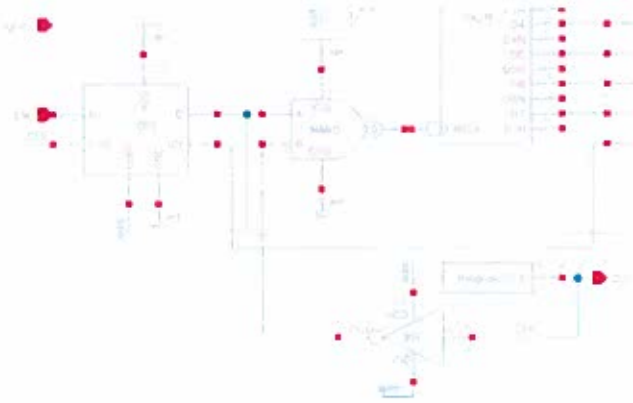


Fig. 3. Shift register clock gating

B. Digital blocks

1) Shift register: The shift register was realised by chaining 8 D-type flip-flops. Special D-type flip-flops that have inverted clock input and both inverting and non-inverting outputs was used, to remove the need of a few inverters.

Fig. 3 shows the clock gating of the shift register. The clock input to the NAND is inverted, to prevent potential clock glitch that might happen when clock rises then buffered enable signal falls a bit later. Because of propagation delay, the clock signal to shift register will be delayed related to the clock generator. Therefore, it would be better to sample the buffered data input at falling edge, hence the inverted clock input.

2) Input reset switches: The input switches are simply 8 NANDs. The input of capacitor array drives MOSFETs arranged like an inverter, thus the output of this stage should be inverted. Inverted enable signal come from input buffer was used, so the output will be high when enable is asserted. Also using the inverted output from shift register, simple NANDs are sufficient.

C. Capacitor array

Fig. 4 shows a 6-bit charge-scaling DAC realised using capacitor array with split capacitor.

After initial discharge, by having binary-weighted capacitors connected to either VREF or GND, the capacitors will be charged, with output node have a voltage proportional to charge distribution, hence the name "charge-scaling". There are no static current flow in the capacitor array, therefore very power efficient. It is very sensitive to parasitic capacitances, so the capacitors need to be big enough to overcome them.

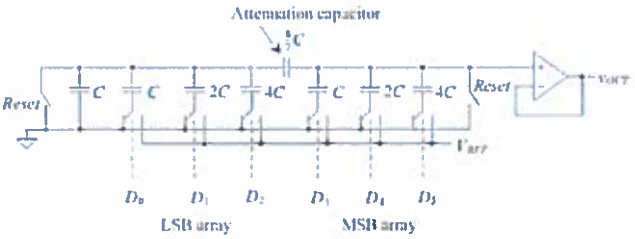


Fig. 4. A charge-scaling DAC using split array (adapted from [3])

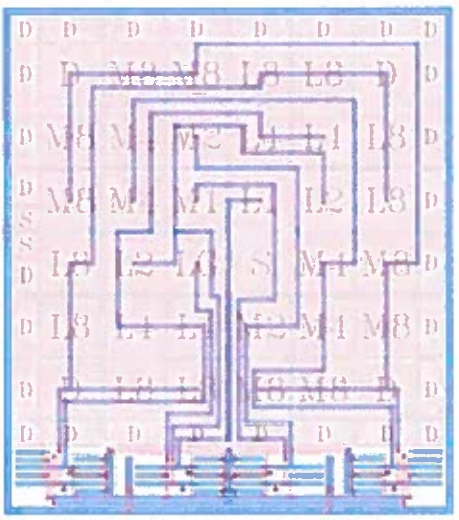


Fig. 5. Lay design of capacitor array

However, if the capacitor array has not been reset for a long time, the output might drift because of leaking current changes charge distribution on output node. Having a larger unit capacitor might slower this process.

The unit capacitor was determined to be $12\mu m \times 12\mu m$, a sensible value for efficient layout area usage and sufficient enough to meet the specifications.

Fig. 5 shows the actual layout design of the capacitor array. M and L capacitors corresponding to capacitors used in most significant bits and least significant bits respectively, while S capacitors form the split capacitor. Dummy capacitors (D) were also added for better matching around the edge.

The capacitors are placed in a common centroid configuration, as indicated by the text on each unit capacitor. In this arrangement, component value variation caused by first order gradient on wafer will be cancelled out.

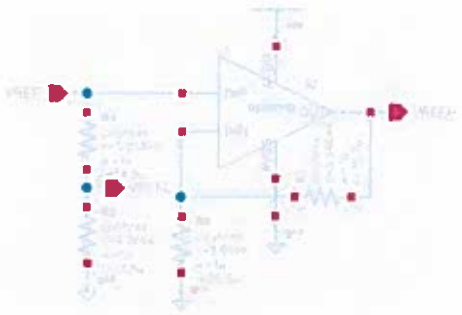


Fig. 6. Schematic design of voltage references

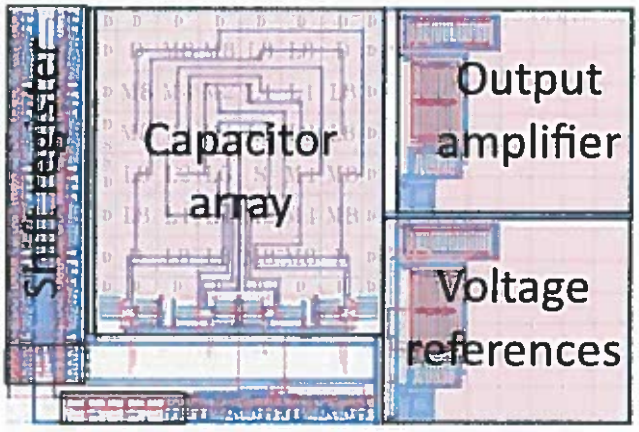


Fig. 7. Layout design of the DAC

D. Voltage references

Fig. 6 shows the schematic of voltage references generation. Since the loads are only capacitors, a simple potential divider should be sufficient for $0.4V$ output. The output will become stable at the desired voltage after capacitors are charged. The $1.4V$ is achieved by using an amplifier, which is a 2-stage op amp design from previous laboratory sessions.

The outputs are tuned a bit higher than the voltage ranges specified, in order to compensate for gain and offset errors, according to simulation.

E. Output amplifier

A simple non-inverting amplifier was used as output amplifier. The same 2-stage op amp was used.

F. Top-level overview

Fig. 7 shows the entire layout design of the DAC. The dimension of the layout is about $200\mu m \times 132\mu m$.

P guarding rings was added around each analogue components to provide better substrate contact and isolation between components. Metal conductors for analogue signals and power tracks were also made thicker, to reduce parasitic resistances.

IV. Simulated Results

The ring oscillator was simulated by applying a $1.8V$ supply. It is simulated and tuned separately, because the simulation of the entire DAC module depends on an accurate clock. Fig. 8 shows the transient simulation waveform of the ring oscillator.

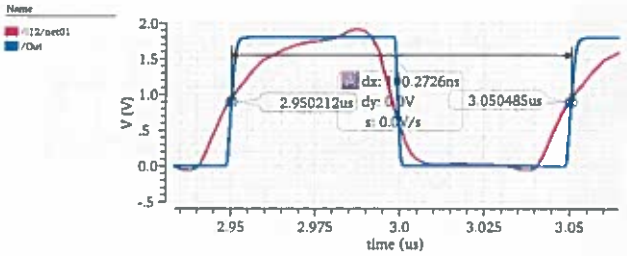


Fig. 8. Transient simulation of ring oscillator. Red: Waveform on the ring; Blue: Buffered output

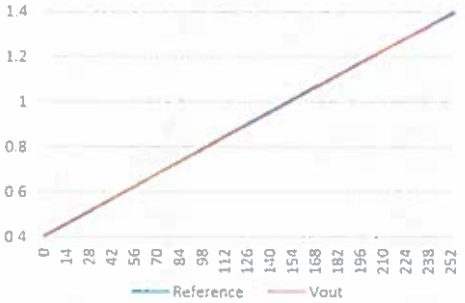


Fig. 9. Reference and simulation output of the DAC design

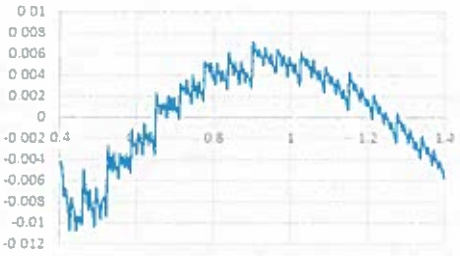


Fig. 10. Relative error of the DAC design output

The frequency of the ring oscillator was calculated as shown in (1). It satisfied the specification of $10 \pm 0.1MHz$.

$$f = \frac{1}{T} \approx \frac{1}{100.2726ns} \approx 9.973MHz \quad (1)$$

Simulation of the DAC module was done by using the provided test script. It will test every input codes, output the results as text. Therefore, the results can be easily processed in a spreadsheet application afterwards.

Fig. 9 shows the reference and simulation output voltages. There is always a small error exists between them.

Fig. 10 shows the relative error between DAC output and reference. Relative error was calculated as shown by (2).

$$ERR_{rel} = abs(Reference - VOUT)/Reference \quad (2)$$

It can be seen from the chart that the overall error is not linear. This is affected by parasitic capacitances on MSB side.

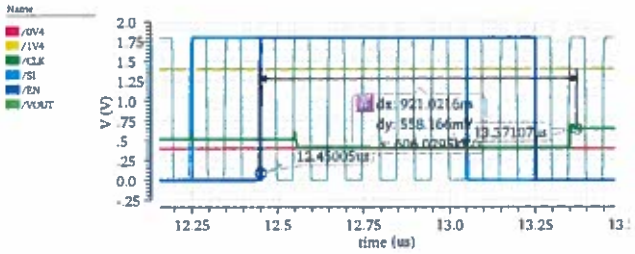


Fig. 11. Transient simulation of the DAC design

TABLE I
TECHNICAL SPECIFICATION

Specification	Value	Unit
Total silicon area	0.0264	mm^2
Power supply	1.8	V
Power dissipation	5.2246	mW
Output clock frequency	9.973	MHz
Conversion rate	1000	kHz
Output voltage range	0.4 - 1.4	V
Load impedance for VOUT	10k // 10	Ω // pF
Relative error	$\leq$ 1.1	%

The parasitic capacitances on LSB side is reflected by small changes in error that repeats every 16 steps.

Voltage references were tuned to move the error curve centred around zero, gives the minimum absolute relative error.

The relative error can be further reduced by increasing the unit capacitance, so that the parasitic capacitances will become more insignificant. However, this will increase the layout area, and since the design already meet the specification of less than 1.5%, the capacitors were not changed.

The test script also gives power consumption values. These can be calculated by measuring the current supplied by the $1.8V$ power source. The static power consumption of this design is $5.2246mW$, and the dynamic power consumption is $5.0504mW$. It satisfied the specification of less than $20mW$.

Fig. 11 shows a fraction of transient simulation for the DAC. It shows the time between the rising edge of enable signal and output stable is about $921ns$, approximately $1.08MHz$, meets the specification of at least $500kHz$.

V. CONCLUSION

The capacitor based charge scaling DAC design successfully meets all design specifications.

TABLE I shows the specification of this DAC design.

REFERENCES

[1] T. G. Constandinou, "Cousework assignment: Full custom design of a digital to analogue converter," 2016.

[2] R. J. Van de Plassche, *CMOS integrated analog-to-digital and digital-to-analog converters*. Springer Science & Business Media, 2013, vol. 742.

[3] T. G. Constandinou, "Lecture 9: Data converters," 2016.

Coursework Assessment

Full Custom Design of a Digital to Analogue Converter (DAC) in 0.18μm CMOS

jz11415

Jan, 2017

CONGRATULATIONS! Your DAC design is functional.

1 Overall Mark

Your Overall Design Mark is A
Your Overall Report Mark is A

2 Design Specifications

Table 1: Achieved Design Specifications

Name	Specifications	Value	Units	Rank	Best	Median
Area	Total silicon area	40800	μm^2	40% - 50%	6300.00	43056.00
P_{Sdiss}	Static Power dissipation	5.674	mW	40% - 50%	1.75	5.67
P_{Ddiss}	Dynamic Power dissipation	5.143	mW	40% - 50%	1.67	5.17
ERR_{rel}	Relative error	0.180	%	0 % - 10%	0.15	0.55
ERR_{clk}	Clock error	-0.010	%	10 % - 20%	0.00	0.42

Best is the best results across 41 students and Median is the median value

3 Comments On The Design

Please find comments for the design submission as below

3.1 Design and functionality

Binary C array based charge redistribution DAC. Unity cap array is used for splitter cap. Active potential devider based reference without buffering could have speed or non-linearity issue.

3.2 Design clarity

Clear top level design and sub cells, with necessary notes. Even custom made some symbols for illustrative purpose.

© 2017 Department of Electrical and Electronic Engineering, Imperial College London

3.3 Layout

Your DAC layout passed DRC
Your DAC layout passed LVS

Layout in a rectangle shape. Dummy Res and resistor matching are not used. Dual row guarding contacts is used between cap array and ring osc. Can add more guard between cap array and digital.

4 Report Evaluation

Please find comments for your report as below

4.1 Design description

Good explanation of the design. The structure adopts first a high level description, then each block is detailed. You should have included the topology for your voltage reference generator, which should also be in a separate section. Schematics very clear, well done. You should have included a timing diagram.

4.2 Circuit Theory

Good background theory on the DAC topology. Some analysis is provided for the capacitor array and the ring oscillator. Table 1 is very good for design comparison, and your justification regarding the capacitor array is much appreciated. Some elements are unclear however, for instance the matching would be quite critical for an R/2R ladder as well, and I am not certain why this topology would be low noise. Overall, your circuit description might have included some more analysis on the transistor sizing and the design decisions.

4.3 Results and interpretation

Very interesting graphs, offset and gain errors are annotated on the figure. Power simulations are carried out, which is a very nice addition to your paper. As a note, keep in mind to consider the power through the 0.9 reference voltage as well, also it would have been interesting to look at the power consumption of each block separately, but these are just suggestions. The performance table is a good summary. Please provide a more accurate value for the sampling frequency.

4.4 Presentation

Good presentation of the paper, matches IEEE standards. The structure is clear, however some figures (especially Figs 6 and 8) need larger labels. Your abstract contains all the information, however make sure you start with a context sentence, follow with a summary of your design and end with the key results. The last sentence is not necessary except for the total chip area.

© 2017 Department of Electrical and Electronic Engineering, Imperial College London

5 Simulation Log

Please find simulation log as below, please note these were the initial simulation results, and were reviewed by markers. Manual resimulated might be performed to get accurate specification for some points

---

This is evaluation results of Full Custom IC Design course work for jz11415
Creating test bench schematics (tcellDACforscript_jz11415 in CMOS16_DAC_jz11415) for design (DAC in CMOS16_DAC_jz11415)
Found terminal CLK of inst INST0
Matched to pin CLK in the assignment
Found terminal VOUT of inst INST0
Matched to pin VOUT in the assignment
Found terminal EN of inst INST0
Matched to pin EN in the assignment
Found terminal SI of inst INST0
Matched to pin SI in the assignment
Found terminal VREF of inst INST0
Matched to pin VREF in the assignment
CLK period is 99.9904 nS, relative error is -0.0096 Static power consumption is 5.6742 mW , dynamic power consumption is 5.1427 mW
Index Reference Vout error
0 0.4039 0.4034 0.0014
1 0.4078 0.4073 0.0014
2 0.4118 0.4112 0.0015
3 0.4157 0.4151 0.0014
4 0.4196 0.4191 0.0013
5 0.4235 0.4230 0.0013
6 0.4275 0.4269 0.0013
7 0.4314 0.4309 0.0012
8 0.4353 0.4348 0.0012
9 0.4392 0.4387 0.0012
10 0.4431 0.4426 0.0012
11 0.4471 0.4466 0.0011
12 0.4510 0.4504 0.0012
13 0.4549 0.4544 0.0011
14 0.4588 0.4584 0.0010
15 0.4627 0.4624 0.0008
16 0.4667 0.4663 0.0008
17 0.4706 0.4701 0.0009
18 0.4745 0.4741 0.0008
19 0.4784 0.4779 0.0010

© 2017 Department of Electrical and Electronic Engineering, Imperial College London

20 0.4824 0.4820 0.0007
21 0.4863 0.4859 0.0007
22 0.4902 0.4899 0.0007
23 0.4941 0.4938 0.0007
24 0.4980 0.4978 0.0005
25 0.5020 0.5017 0.0006
26 0.5059 0.5056 0.0005
27 0.5098 0.5095 0.0005
28 0.5137 0.5135 0.0004
29 0.5176 0.5174 0.0005
30 0.5216 0.5214 0.0004
31 0.5255 0.5252 0.0005
32 0.5294 0.5293 0.0002
33 0.5333 0.5331 0.0004
34 0.5373 0.5372 0.0001
35 0.5412 0.5411 0.0002
36 0.5451 0.5451 0.0000
37 0.5490 0.5490 0.0001
38 0.5529 0.5529 0.0000
39 0.5569 0.5568 0.0001
40 0.5608 0.5608 0.0001
41 0.5647 0.5647 0.0000
42 0.5686 0.5687 0.0001
43 0.5725 0.5726 0.0001
44 0.5765 0.5766 0.0002
45 0.5804 0.5805 0.0002
46 0.5843 0.5844 0.0002
47 0.5882 0.5885 0.0004
48 0.5922 0.5923 0.0003
49 0.5961 0.5963 0.0003
50 0.6000 0.6002 0.0003
51 0.6039 0.6042 0.0005
52 0.6078 0.6081 0.0004
53 0.6118 0.6120 0.0005
54 0.6157 0.6160 0.0005
55 0.6196 0.6199 0.0005
56 0.6235 0.6239 0.0006
57 0.6275 0.6278 0.0006
58 0.6314 0.6318 0.0007
59 0.6353 0.6357 0.0007
60 0.6392 0.6397 0.0007
61 0.6431 0.6436 0.0007

© 2017 Department of Electrical and Electronic Engineering, Imperial College London

62 0.6471 0.6475 0.0007
63 0.6510 0.6515 0.0007
64 0.6549 0.6553 0.0007
65 0.6588 0.6593 0.0008
66 0.6627 0.6633 0.0008
67 0.6667 0.6672 0.0009
68 0.6706 0.6712 0.0009
69 0.6745 0.6751 0.0008
70 0.6784 0.6790 0.0009
71 0.6824 0.6830 0.0009
72 0.6863 0.6870 0.0010
73 0.6902 0.6909 0.0010
74 0.6941 0.6948 0.0010
75 0.6980 0.6987 0.0009
76 0.7020 0.7027 0.0011
77 0.7059 0.7066 0.0010
78 0.7098 0.7106 0.0011
79 0.7137 0.7145 0.0011
80 0.7176 0.7185 0.0012
81 0.7216 0.7224 0.0011
82 0.7255 0.7264 0.0013
83 0.7294 0.7303 0.0013
84 0.7333 0.7342 0.0012
85 0.7373 0.7382 0.0013
86 0.7412 0.7422 0.0013
87 0.7451 0.7460 0.0013
88 0.7490 0.7500 0.0013
89 0.7529 0.7539 0.0013
90 0.7569 0.7579 0.0014
91 0.7608 0.7618 0.0014
92 0.7647 0.7657 0.0013
93 0.7686 0.7697 0.0014
94 0.7725 0.7735 0.0013
95 0.7765 0.7776 0.0014
96 0.7804 0.7815 0.0014
97 0.7843 0.7854 0.0014
98 0.7882 0.7894 0.0015
99 0.7922 0.7933 0.0014
100 0.7961 0.7973 0.0015
101 0.8000 0.8012 0.0015
102 0.8039 0.8052 0.0015
103 0.8078 0.8091 0.0015

© 2017 Department of Electrical and Electronic Engineering, Imperial College London

104 0.8118 0.8130 0.0016
105 0.8157 0.8169 0.0015
106 0.8196 0.8209 0.0016
107 0.8235 0.8248 0.0016
108 0.8275 0.8288 0.0016
109 0.8314 0.8327 0.0016
110 0.8353 0.8366 0.0016
111 0.8392 0.8405 0.0016
112 0.8431 0.8445 0.0016
113 0.8471 0.8484 0.0016
114 0.8510 0.8523 0.0016
115 0.8549 0.8563 0.0016
116 0.8588 0.8602 0.0016
117 0.8627 0.8642 0.0017
118 0.8667 0.8680 0.0016
119 0.8706 0.8720 0.0016
120 0.8745 0.8759 0.0016
121 0.8784 0.8799 0.0016
122 0.8824 0.8837 0.0016
123 0.8863 0.8877 0.0016
124 0.8902 0.8916 0.0016
125 0.8941 0.8956 0.0016
126 0.8980 0.8995 0.0016
127 0.9020 0.9034 0.0016
128 0.9059 0.9073 0.0016
129 0.9098 0.9113 0.0016
130 0.9137 0.9152 0.0016
131 0.9176 0.9191 0.0016
132 0.9216 0.9231 0.0016
133 0.9255 0.9269 0.0016
134 0.9294 0.9309 0.0016
135 0.9333 0.9348 0.0015
136 0.9373 0.9387 0.0016
137 0.9412 0.9426 0.0016
138 0.9451 0.9466 0.0016
139 0.9490 0.9505 0.0015
140 0.9529 0.9544 0.0016
141 0.9569 0.9583 0.0015
142 0.9608 0.9623 0.0016
143 0.9647 0.9662 0.0015
144 0.9686 0.9701 0.0016
145 0.9725 0.9741 0.0016

© 2017 Department of Electrical and Electronic Engineering, Imperial College London

146 0.9765 0.9780 0.0016
147 0.9804 0.9819 0.0016
148 0.9843 0.9858 0.0016
149 0.9882 0.9897 0.0015
150 0.9922 0.9936 0.0014
151 0.9961 0.9975 0.0014
152 1.0000 1.0015 0.0015
153 1.0039 1.0053 0.0014
154 1.0078 1.0093 0.0014
155 1.0118 1.0132 0.0014
156 1.0157 1.0171 0.0014
157 1.0196 1.0210 0.0014
158 1.0235 1.0249 0.0013
159 1.0275 1.0288 0.0013
160 1.0314 1.0328 0.0014
161 1.0353 1.0367 0.0014
162 1.0392 1.0405 0.0013
163 1.0431 1.0445 0.0013
164 1.0471 1.0484 0.0013
165 1.0510 1.0523 0.0013
166 1.0549 1.0562 0.0013
167 1.0588 1.0601 0.0012
168 1.0627 1.0640 0.0012
169 1.0667 1.0679 0.0011
170 1.0706 1.0718 0.0012
171 1.0745 1.0757 0.0011
172 1.0784 1.0796 0.0011
173 1.0824 1.0835 0.0011
174 1.0863 1.0875 0.0011
175 1.0902 1.0913 0.0010
176 1.0941 1.0953 0.0010
177 1.0980 1.0992 0.0010
178 1.1020 1.1030 0.0009
179 1.1059 1.1069 0.0009
180 1.1098 1.1109 0.0010
181 1.1137 1.1147 0.0009
182 1.1176 1.1185 0.0008
183 1.1216 1.1225 0.0008
184 1.1255 1.1263 0.0007
185 1.1294 1.1303 0.0008
186 1.1333 1.1341 0.0007
187 1.1373 1.1381 0.0007

© 2017 Department of Electrical and Electronic Engineering, Imperial College London

188 1.1412 1.1420 0.0007
189 1.1451 1.1459 0.0007
190 1.1490 1.1497 0.0006
191 1.1529 1.1536 0.0006
192 1.1569 1.1575 0.0005
193 1.1608 1.1614 0.0005
194 1.1647 1.1653 0.0005
195 1.1686 1.1691 0.0004
196 1.1725 1.1731 0.0005
197 1.1765 1.1770 0.0004
198 1.1804 1.1809 0.0004
199 1.1843 1.1847 0.0003
200 1.1882 1.1886 0.0003
201 1.1922 1.1925 0.0003
202 1.1961 1.1964 0.0003
203 1.2000 1.2002 0.0002
204 1.2039 1.2041 0.0002
205 1.2078 1.2080 0.0001
206 1.2118 1.2118 0.0001
207 1.2157 1.2158 0.0001
208 1.2196 1.2196 0.0000
209 1.2235 1.2235 0.0000
210 1.2275 1.2274 0.0000
211 1.2314 1.2313 0.0001
212 1.2353 1.2351 0.0001
213 1.2392 1.2389 0.0002
214 1.2431 1.2429 0.0002
215 1.2471 1.2467 0.0003
216 1.2510 1.2506 0.0003
217 1.2549 1.2544 0.0004
218 1.2588 1.2584 0.0004
219 1.2627 1.2622 0.0004
220 1.2667 1.2661 0.0005
221 1.2706 1.2700 0.0005
222 1.2745 1.2739 0.0005
223 1.2784 1.2777 0.0006
224 1.2824 1.2815 0.0007
225 1.2863 1.2854 0.0007
226 1.2902 1.2893 0.0007
227 1.2941 1.2931 0.0008
228 1.2980 1.2970 0.0008
229 1.3020 1.3008 0.0009

© 2017 Department of Electrical and Electronic Engineering, Imperial College London

230 1.3059 1.3047 0.0009
231 1.3098 1.3086 0.0009
232 1.3137 1.3124 0.0010
233 1.3176 1.3162 0.0011
234 1.3216 1.3201 0.0011
235 1.3255 1.3240 0.0011
236 1.3294 1.3279 0.0012
237 1.3333 1.3317 0.0012
238 1.3373 1.3356 0.0012
239 1.3412 1.3395 0.0013
240 1.3451 1.3433 0.0013
241 1.3490 1.3472 0.0013
242 1.3529 1.3511 0.0014
243 1.3569 1.3549 0.0015
244 1.3608 1.3588 0.0014
245 1.3647 1.3626 0.0015
246 1.3686 1.3665 0.0016
247 1.3725 1.3704 0.0016
248 1.3765 1.3742 0.0016
249 1.3804 1.3781 0.0017
250 1.3843 1.3820 0.0017
251 1.3882 1.3858 0.0017
252 1.3922 1.3897 0.0018
253 1.3961 1.3936 0.0018
254 1.4000 1.3974 0.0018
Max ERROR is 0.0018
Simulation finished

© 2017 Department of Electrical and Electronic Engineering, Imperial College London

Laboratory 2 Inverter Layout & Digital Cells

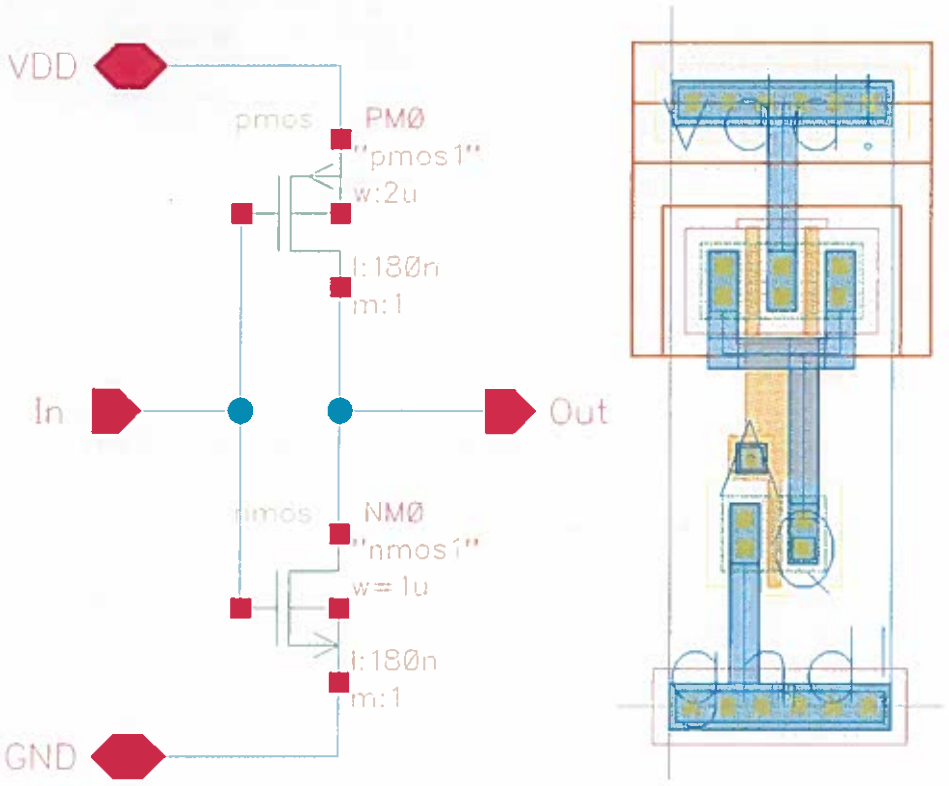
Yubo Zhi (yz4116)

October 27, 2016

1 Inverter

1.1 Schematic

Figure 1a shows the schematic of a inverter cell, including parameters for MOSFETs, designed from the previous laboratory session.



(a) Schematic including MOSFET parameters (b) Final layout design

Figure 1: A inverter cell

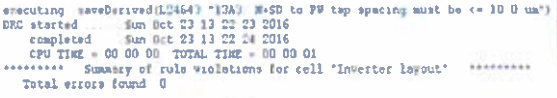
1.2 Layout

Figure 1b shows the final layout designed for the inverter. The P-channel MOSFET (PMOS) is split into 2 fingers, to meet the height requirement for the standard cells ($8\mu m$ in this case). The N-channel MOSFET (NMOS) and P-channel MOSFET (PMOS) are moved closer to the vertical

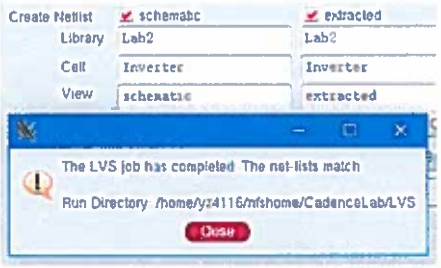
centre, and VDD is supplied to the centre tap of the PMOS instead, compare to the example layout given in laboratory notes. This is to accommodate the routing requirements at top and bottom spaces, used by the D-type Flip-Flop designed later.

To achieve minimum cell width, the spacing between cell boundaries to the left and right of the P+ implant inside the inverter PMOS should add up to the minimum value defined by the design rules, and kept the same across all standard cells. Therefore, when two standard cells are tightly placed next to each other, the design rule will not be violated, and no space will be wasted. The gpdk180 design rules shows the requirement is $0.4\mu m$. Therefore, the spacing is determined to be $0.2\mu m$ between the P+ implant and the cell boundaries.

The DRC and LVS verification were then executed, as shown in Figure 2. No DRC errors ensures the layout designed does not violate any design rules. LVS verification ensures the layout matches the schematic and parameters.



(a) DRC checking result, no errors



(b) LVS checking result, the extracted net-list matches the schematic

Figure 2: Inverter cell layout verification

2 NAND gate

2.1 Schematic

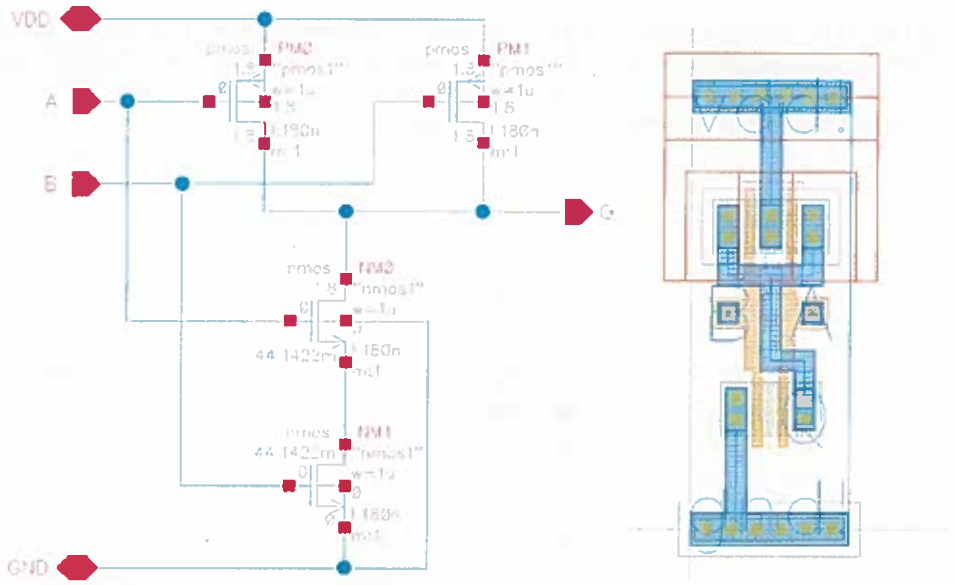
Figure 3a shows the schematic of a NAND cell, including parameters for MOSFETs.

2.2 Determining MOSFET parameters

According to specification, the NAND gate should be about to drive a $1pF$ capacitor with a maximum propagation delay of $10ns$. By applying the time constant equation (Equation 1), the resistance for charging the capacitive load should be no more than $10k\Omega$.

$$\tau = RC \tag{1}$$
$$R = \frac{\tau}{C} = \frac{10ns}{1pF} = 10k\Omega$$

The ON resistances ratio of PMOS and NMOS need to be determined first for transistor sizing. The BSIM3V3 MOSFET simulation models used by the simulator are too complex for direct analysis. Therefore, a simulation of the inverter was used to extract these parameters, as shown in Figure 4. The input of the inverter is a square wave with period of $1\mu s$. The operating points of the circuit were saved every $0.5\mu s$ after $0.25\mu s$ into the simulation, when the inverter output is stable. This was done by specifying the logging times in the `infotime` parameter in the transient simulation options.



(a) Schematic including MOSFET parameters (b) Final layout design

Figure 3: A NAND cell

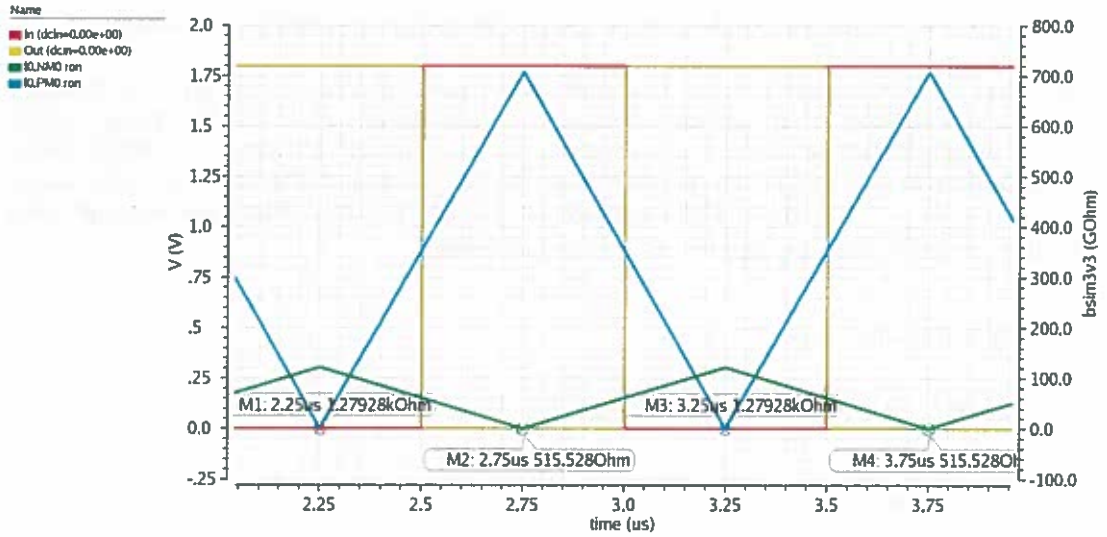


Figure 4: Transient simulation of a Inverter cell with operating points logging

From the results browser, the ron parameters logged can be plotted with the input and output waveforms. The markers on Figure 4 shows the ON resistances of PMOS and NMOS transistors, about $1.28k\Omega$ and 516Ω respectively. The widths of PMOS and NMOS simulated are $2\mu m$ and $1\mu m$ respectively. According to MOSFET modelling (Equation 2), the ON resistance should be inversely proportional to the width of MOSFETs.

$$g_m = \mu_n C_{ox} \frac{W}{L}(V_{GS} - V_{TH}) \approx \frac{1}{R_{on}} \tag{2}$$

For balancing the NAND cell, the ON resistances of the PMOS branch and the NMOS branch should be the same. This can be evaluated when both inputs are changing, e.g. from 00 to 11 and

from 11 to 00. The output switching point when this is happening should be at half the supply voltage ($1.8V$), which is about $900mV$. At this switching point, all PMOS and NMOS transistors are on. By modelling the PMOS and NMOS transistors as resistors of their own ON resistances, the circuit becomes a potential divider, as shown in Figure 5.

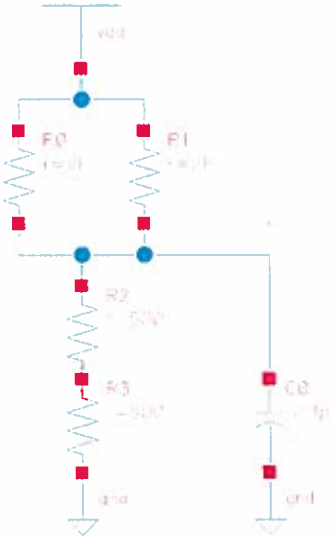


Figure 5: NAND cell switching point analysis, by modelling MOSFETs as resistors

The widths of NMOS transistors are determined to be $1\mu m$, the same as the inverter. So the 2 series connected NMOS should add up to a resistance of about $516\Omega \times 2 = 1.032k\Omega$. Using a width of $1\mu m$, a single PMOS will have an ON resistance of approximately $1.28k\Omega \times 2 = 2.56k\Omega$. The 2 PMOS transistors are connected in parallel, thus the total resistance of the PMOS branch would be about $\frac{1}{\frac{1}{2.56k} + \frac{1}{2.56k}}\Omega = 1.28k\Omega$. This value is roughly matched with the NMOS branch, and much less than the $10k\Omega$ value required by the timing specification.

2.3 Simulation

Figure 6 shows the test bench set up for simulating the NAND cell.

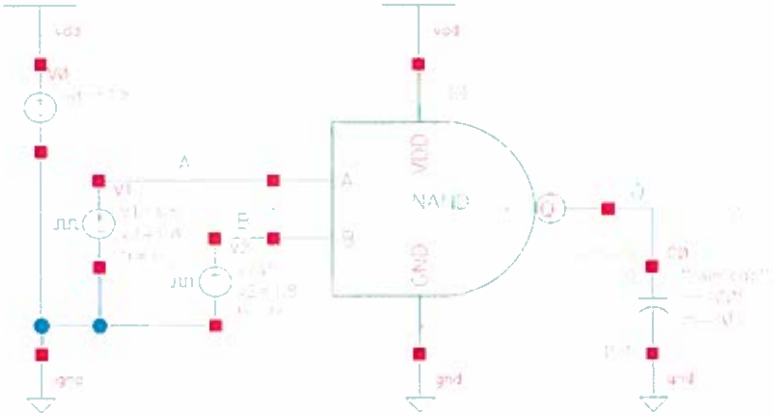
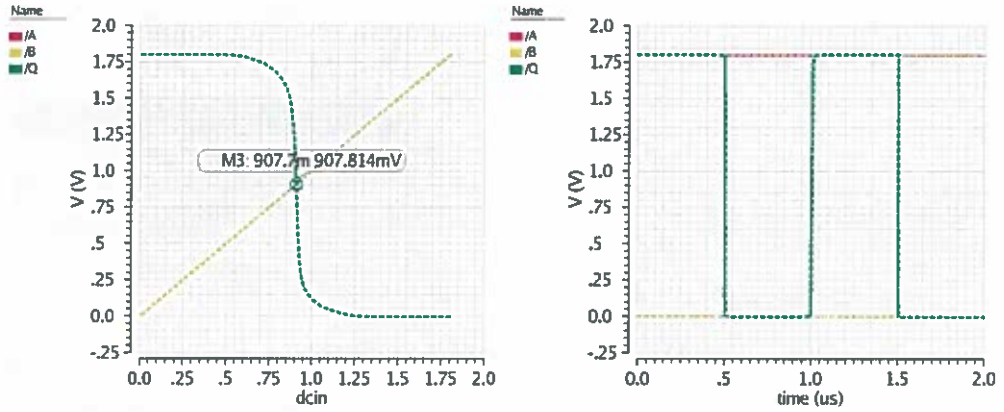


Figure 6: NAND simulation test bench set up

Figure 7 shows the simulation results for DC sweep and transient analysis.

For DC sweep analysis, both inputs are changing from $0V$ to $1.8V$. As the marker suggests, the

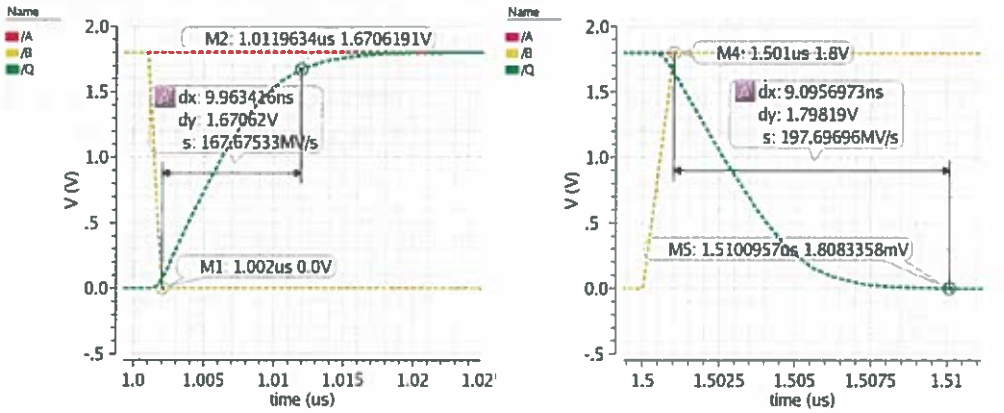


(a) DC sweep (b) Transient

Figure 7: NAND cell simulation

switching point is about $908mV$, approximately at half of the power supply. Thus the transistor sizing proposed above do result in a balanced design.

For transient simulation, input A is kept at logic high, only input B is toggling. This gives the worst case propagation delay when output changes from 0 to 1, where only 1 PMOS transistor will be on for driving the output to the required logic level. The close-up of output edges are shown in Figure 8. For output rising edge, the output rises to about $1.67V$ or 93% in $10ns$, can be considered within specification.



(a) Output rising edge (b) Output falling edge

Figure 8: Close-up of transient simulation

2.4 Layout

Figure 3b shows the final layout designed for the NAND cell. Spacing between P+ implant of the PMOS and cell boundaries is $0.2\mu m$, as determined earlier during inverter layout design, gives the minimum possible cell width. The 2 PMOS transistors share a common VDD tap, thus can be combined together. The 2 series connected NMOS transistors can also be combined, with no central tap needed. These combinations reduce the cell width a lot, and the layout now looks quite similar to the inverter layout.

This layout design passed the DRC and LVS verification, as shown in Figure 9.



(a) DRC checking result, no errors

(b) LVS checking result, the extracted netlist matches the schematic

Figure 9: NAND cell layout verification

3 Edge triggered D Flip-Flop

3.1 Schematic

Figure 12 shows the schematic and MOSFET parameters of the D Flip-Flop (DFF).

The transmission gate MOSFETs are sized as $2\mu m$ for PMOS and $1\mu m$ for NMOS, the same as the inverter.

3.2 Simulation

The DFF cell was simulated using the test bench schematic in laboratory notes. The inputs are set up to show all possible operations a DFF may do, including asynchronous resetting and latching both high and low level signals.

Figure 10 shows the transient simulation result. The waveform clearly shows the propagation delay between clock rising edge and a stable output is much less than $10\mu s$, as required by $100MHz$ operation.

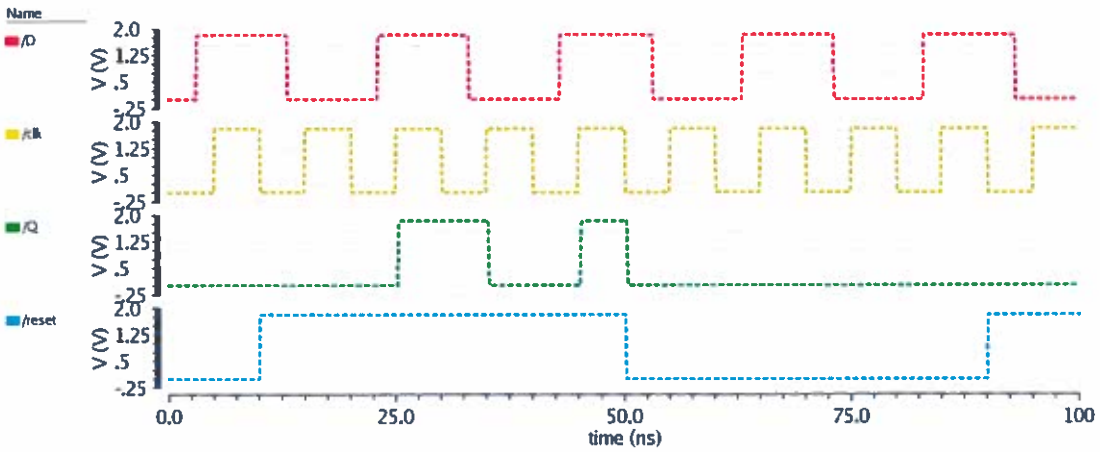


Figure 10: Transient simulation of the DFF design

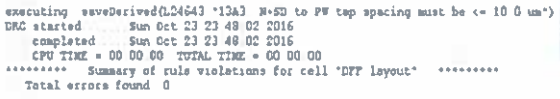
3.3 Layout

Figure 13 shows the final layout design for the DFF cell. For reducing cell width, the standard cells are placed tightly next to each other, and the MOSFETs of the 4 transmission gates were combined

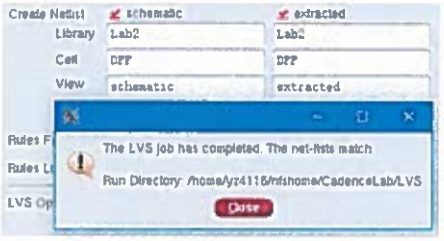
as 2 groups. Several iterations, involving modifying the inverter and NAND layout design, reducing transmission gate PMOS transistors from 2 fingers to 1 finger, were made to achieve minimum cell width. The final cell size achieved was $26.64\mu m \times 8\mu m$.

While keeping the cell as compact as possible, the routings inside the cell were designed as thick as possible, especially for poly silicon layer routings. The sheet resistance for poly silicon is much higher than metal layer, therefore poly silicon routings should be kept as short and thick as possible. The gpdk180 reference manual states the sheet resistances for ploy and metal are 7.5 and 0.1 Ω/square respectively, it is very easy for a long thin ploy silicon trace to reach several kilo ohms of resistance. For this purpose, the VDD tap for the inverter was changed to the centre instead of 2 outer taps as in the example layout.

This layout design passed the DRC and LVS verification, as shown in Figure 11.



(a) DRC checking result, no errors



(b) LVS checking result, the extracted net-list matches the schematic

Figure 11: DFF cell layout verification

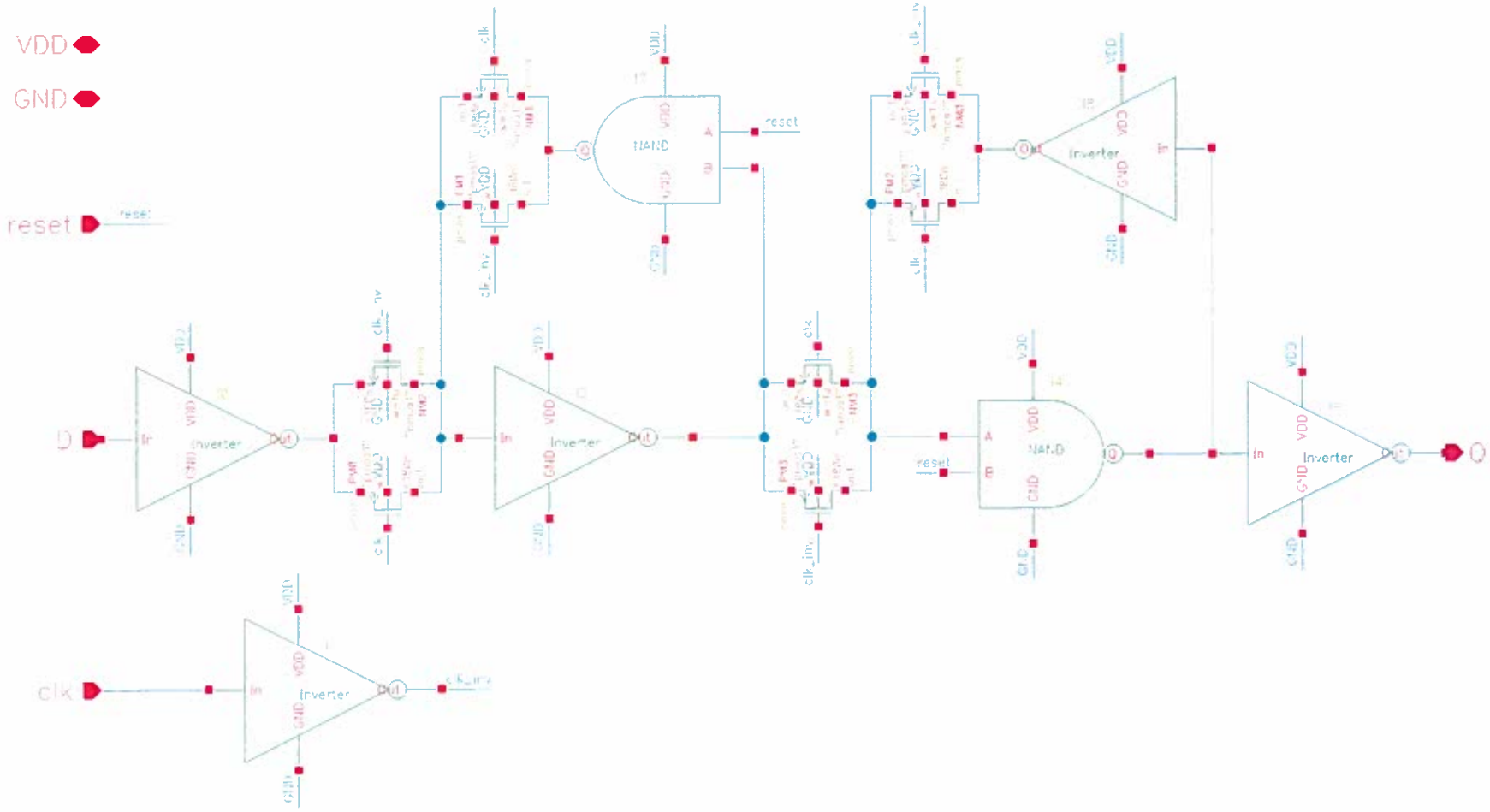


Figure 12: Schematic of DFF including MOSFET parameters

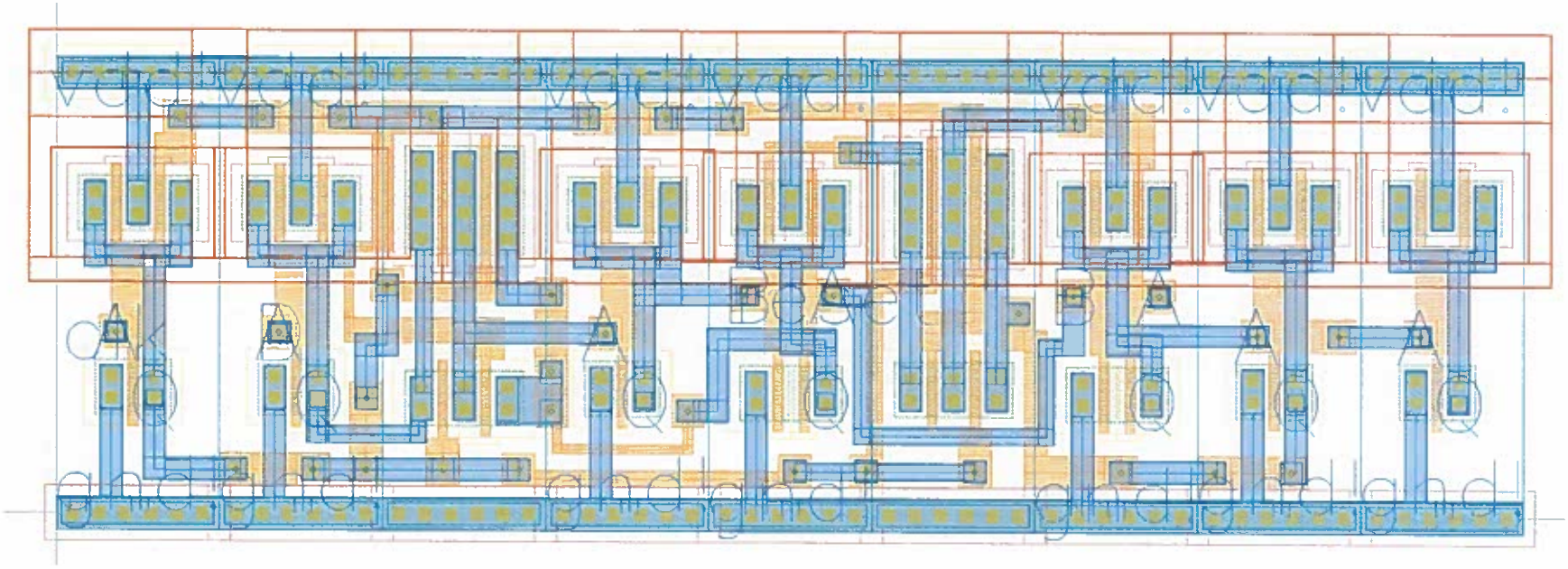


Figure 13: Final layout design of DFF cell

Lab 2 logbook submission_yz4116_attempt_2016-10-27-23-59-41
Name: Norman Zhi (yz4116)
Assignment: Lab 2 logbook submission
Date Submitted: Thursday, 27 October 2016 23:59:41 o'clock BST
Current Grade: 75.0

Submission Field:
There is no student submission text data for this assignment.

Comments:
There are no student comments for this assignment.

Files:
Original filename: logbook.pdf
Filename: Lab 2 logbook submission_yz4116_attempt_2016-10-27-23-59-41_logbook.pdf

Laboratory 3 Analogue Design Flow

Yubo Zhi (yz4116)

4th November 2016

1 Schematic

The objective of this laboratory is to layout a two stage Miller OTA.

Figure 1 shows the schematic of the OTA, including parameters for MOSFETs.

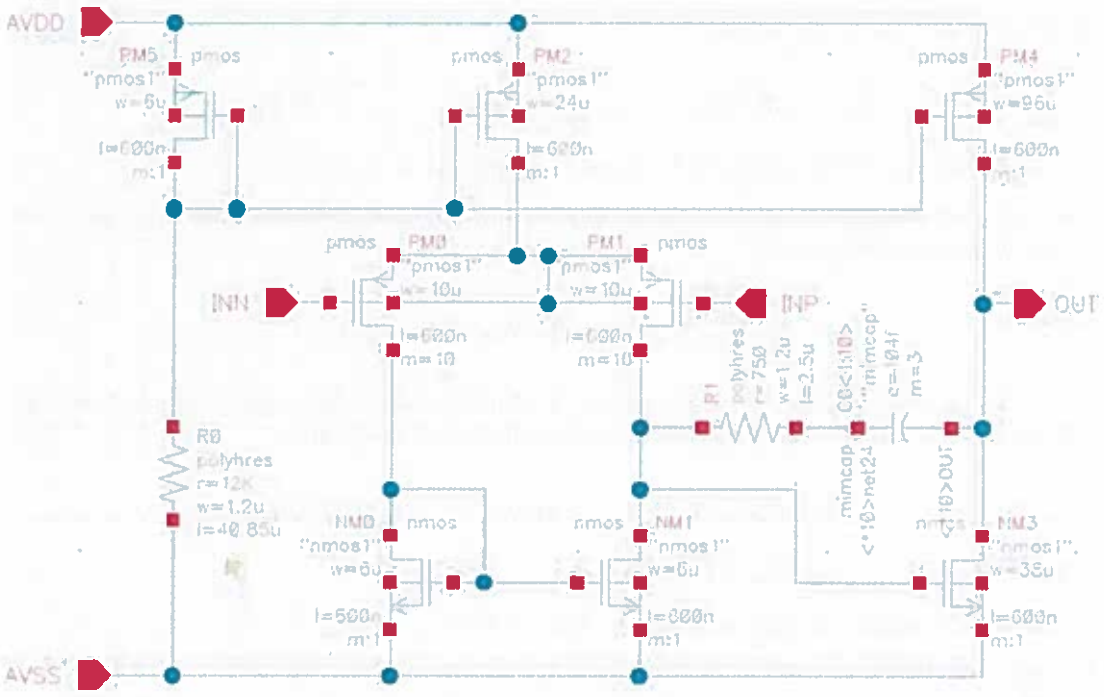


Figure 1: Schematic including MOSFET parameters

PM0 and PM1, the 2 PMOS transistors that forms the differential pair, were changed to a multiplication of 10 MOSFETs with channel width of $10\mu m$ instead of $100\mu m$ as in the notes, in order for better device matching between them using common centroid technique.

PM2 and PM4, the 2 PMOS transistors used in a current mirror configuration, were changed to use a base width of $6\mu m$ with 4 and 16 fingers respectively. This is to match the other transistor PM5 used in the same current mirror configuration, which has a width of $6\mu m$.

NM3, the common source NMOS amplifier, was changed to 8 fingers of width $4.75\mu m$, to improve layout area efficiency. This also reduces gate resistance, compared with a MOSFET with very long gate width.

2 Characterisation

The simulation test bench circuit and set up used in this section were the same as the notes suggests.

2.1 Gain

This op amp consists of two gain stages, a differential input stage and an amplification stage.

The differential gain of the long tailed pair (LTP) differential stage can be modelled as:

$$A_1 = \frac{V_o}{V_{inp} - V_{inn}} = \frac{I_o R_o}{V_d} = \frac{2g_{mp} \times \frac{V_d}{2}}{V_d}(r_{on1} \parallel r_{op1}) = g_{mp}(r_{on1} \parallel r_{op1})$$

Where g_{mp} refers to the transconductance of the differential pair PMOS 0 and 1, and r_{on1} and r_{op1} are the output resistances of NMOS1 and PMOS1 respectively.

Using results browser and DC operating points, the g_m and r_o parameters of these MOSFETs under saturation can be extracted:

$$g_{pm} \approx 1.323mS; \qquad r_{on1} \approx 148.5k\Omega; \qquad r_{op1} \approx 110.4k\Omega$$

The gain of the differential stage can be therefore calculated as $A_1 \approx 83.78$.

The second amplification stage is a common source amplifier, using a single NMOS transistor. The gain of this stage is therefore:

$$A_2 = \frac{V_o}{V_i} = \frac{I_o R_o}{V_i} = g_{mn}(R_{on3} \parallel R_{op4} \parallel R_L)$$

Where R_{on3} and R_{op4} is the output resistances of NMOS3 and PMOS4 respectively, and the load resistance is $R_L = 10k\Omega$. From DC operating point data under saturation:

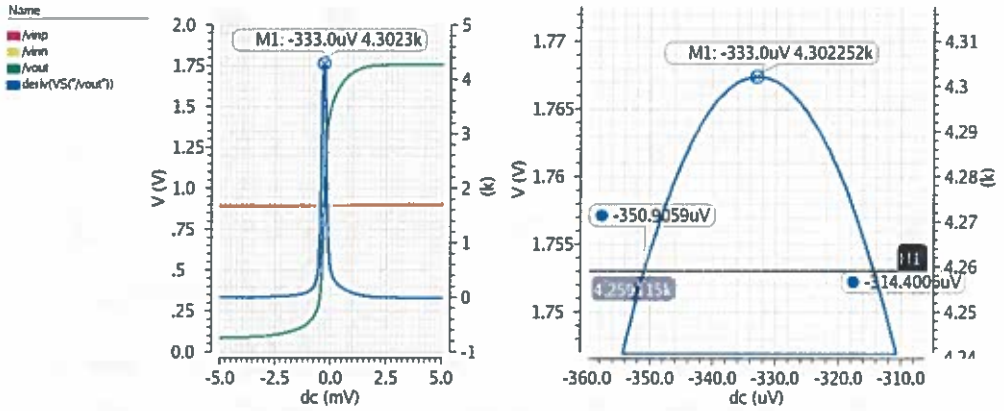
$$g_{mn} \approx 5.371mS; \qquad R_{on3} \approx 19.82k\Omega; \qquad R_{op4} \approx 16.50k\Omega$$

So the gain of the amplification stage is $A_2 \approx 25.45$.

Therefore, the overall voltage gain is $A_{OL} = A_1 A_2 \approx 2132$.

This value can then be verified by plotting V_{OUT} against $V_{INP} - V_{INN} = V_d$. For tiny changes of V_d around a specific operating point, the variation of V_{OUT} may be considered as a linear variation, i.e. $A = \frac{\delta V_{OUT}}{\delta V_d}$. Therefore, a DC sweep analysis can be used to evaluate the open loop gain, by plotting the gradient of the transfer curve, then find the maximum value. This can be done by using the `deriv` calculator function. As shown in Figure 2b, the open loop gain is about $4.302k/2 = 2.151k$, matches the calculated value. The value was divided by 2 because the horizontal axis of the DC sweep analysis is half the differential voltage input.

As can be seen from the DC sweep analysis, the maxima on the gain curve has a slight offset against the $0V$ differential voltage. This is mostly due to mismatch of gain and output resistance values of the bottom current mirror and the amplification stage (PMOS4 and NMOS3), resulting in a small input offset voltage. However, the Virtuoso suite only saves operating point when the differential voltage is exactly $0V$. A small offset voltage of $660\mu V$ was therefore applied to the `INN` terminal, moving the maxima towards the origin, before extracting MOSFET parameters. This improved the accuracy of gain calculation a lot.



(a) Overview (b) Peak open loop gain

Figure 2: DC analysis of the OTA design

2.2 Cut-off frequency

Figure 3 shows a simplified small signal model of the op amp with compensation using Miller effect.

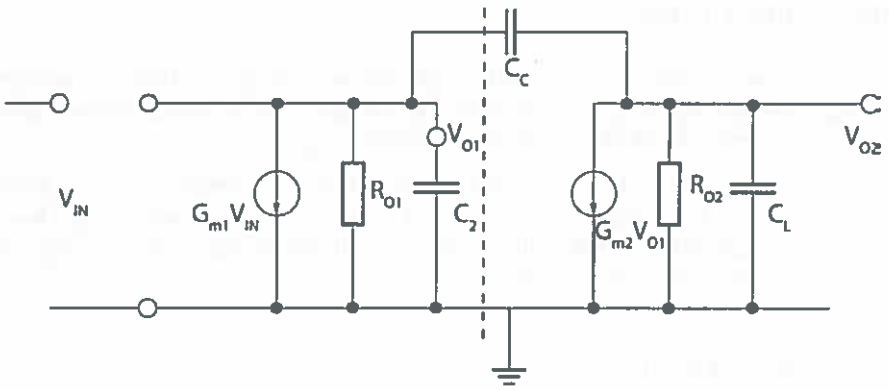


Figure 3: Simplified small signal model of an op amp with compensation

Applying Miller's approximation, the frequency of the dominate pole at low frequency is given by:

$$fp_1 = \frac{1}{2\pi G_{m2} R_{o1} R_{o2} C_c}$$

Where G_{m2} is g_m of NMOS3, R_{o1} is $R_{op1} \parallel R_{on1}$, R_{o2} is $R_{on3} \parallel R_{op3} \parallel R_L$, and $C_c = 10 \times 3 \times 104 fF$.

Using the numbers from operating point extraction and schematic design, the result is $fp_1 = f_{3dB} \approx 30.22kHz$.

Together with the open loop gain, these can be used to derive the bandwidth of this op amp, by using the gain-bandwidth product, where

$$GB = A \times f$$

Therefore, the bandwidth when gain = 0 should be approximately

$$f_{bw} = \frac{GB}{1} = A_{OL} f_{3dB} \approx 65.0MHz$$

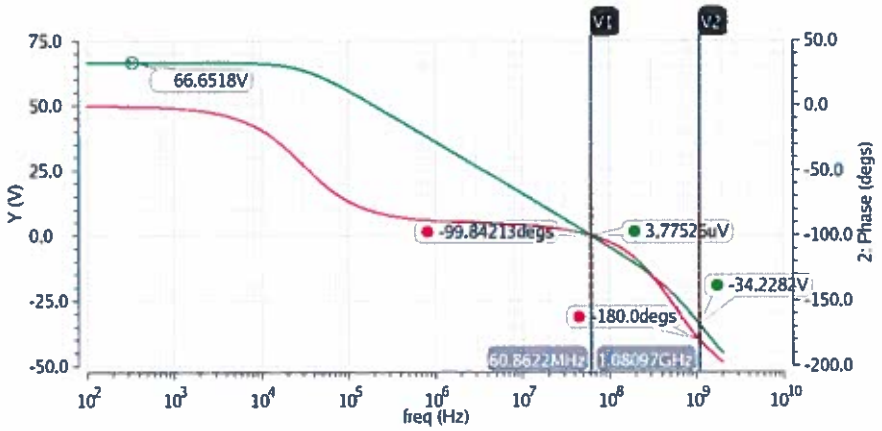


Figure 4: AC analysis of the op amp

By doing an AC analysis with the loads, the bandwidth can be evaluated as shown in Figure 4.

The bandwidth value from simulation is about $60.9MHz$ at marker V1, matches the calculation.

2.3 Linear input range

The linear input range is defined as the range when transconductance of the op amp reduced by 1%. Assuming constant output resistance, this will result in the open loop voltage gain reduced by 1%, which is shown in the DC sweep analysis (Figure 2b).

As the horizontal axis is half the differential voltage, the linear range should be from $-350.9\mu V \times 2 = -701.8\mu V$ to $314.4\mu V \times 2 = -628.8\mu V$, about $73\mu V$. This is a very limited range, but since the op amp has a very large gain, and typically works in a feedback configuration, having a non-linear input characteristic will not cause a significant problem.

2.4 Gain improvements

The overall gain of this op amp can be improved by increasing the transconductance g_m of some transistors, including PMOS0 and PMOS1 in the differential pair configuration, NMOS3 and PMOS4 in the amplification stage.

According to MOSFET modelling equation (Equation 1), for a given technology (i.e. constant doping concentration, gate oxide capacitance, etc.), the transconductance may be improved by increasing $\frac{W}{L}$ ratio. For analogue design, it is preferred to increase the gate length, but not decrease gate width, to avoid device mismatch issues. Increasing the supply voltage, which will in turn increase V_{GS}, may also increase the gain of the amplifier.

$$g_m = \mu_n C_{ox} \frac{W}{L}(V_{GS} - V_{TH}) \tag{1}$$

2.5 Slew rate

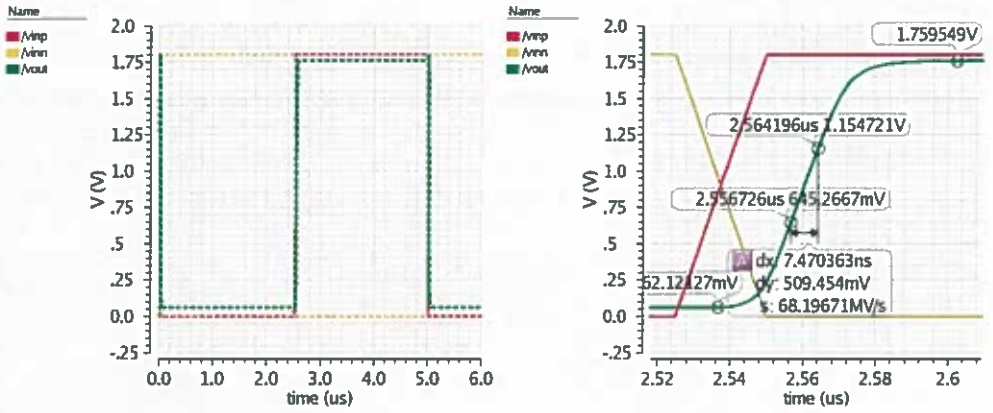
Slew rate is defined as the maximum output swing speed the amplifier can achieve, i.e. $SR = \max(\frac{\Delta V}{\Delta t})$.

For this op amp, it can be evaluated as:

$$SR = \max(\frac{\Delta V}{\Delta t}) =< max(\frac{i_c}{C_L})$$

Biasing current determines the maximum current (i_c) the amplifier can output to drive capacitive loads (C_L). It is proportional to the slew rate, according to the equation. The biasing currents for PMOS4 and NMOS3 determines the slew rate for output rising and falling respectively.

Figure 5b shows a rising edge of the amplifier output from transient analysis. The delta marker shows the gradient of the rising output, about $68.2MV/s$, is the slew rate of the op amp.



(a) Overview

(b) Output rising edge

Figure 5: Transient analysis of the OTA design

3 Layout

Figure 6 shows the layout designed for the OTA.

This layout design passed the DRC and LVS verification, as shown in Figure 7.

For better device matching, the PMOS transistors PM0 and PM1 from differential pair were placed closer together, using common centroid layout technique, as shown in Figure 8. These 2 MOSFETs were split to 10 smaller transistors each, represented by the red and green boxes respectively.

4 dummy poly silicon gates including active area were added to the edges of the LTP. These reduce the edge effect due to etching, improves device matching.

The current mirror PM5, PM2 and PM4 were placed together and implemented using a base width of $6\mu m$, and multiple fingers for multiplication. This gives better multiplication ratio with identical unit devices. Figure 9 shows the layout details of the current mirror. Dummy poly silicon gates were also added to reduce edge effect.

NM3 from amplification stage was also split into multiple fingers, to reduce parasitic resistances and capacitances.

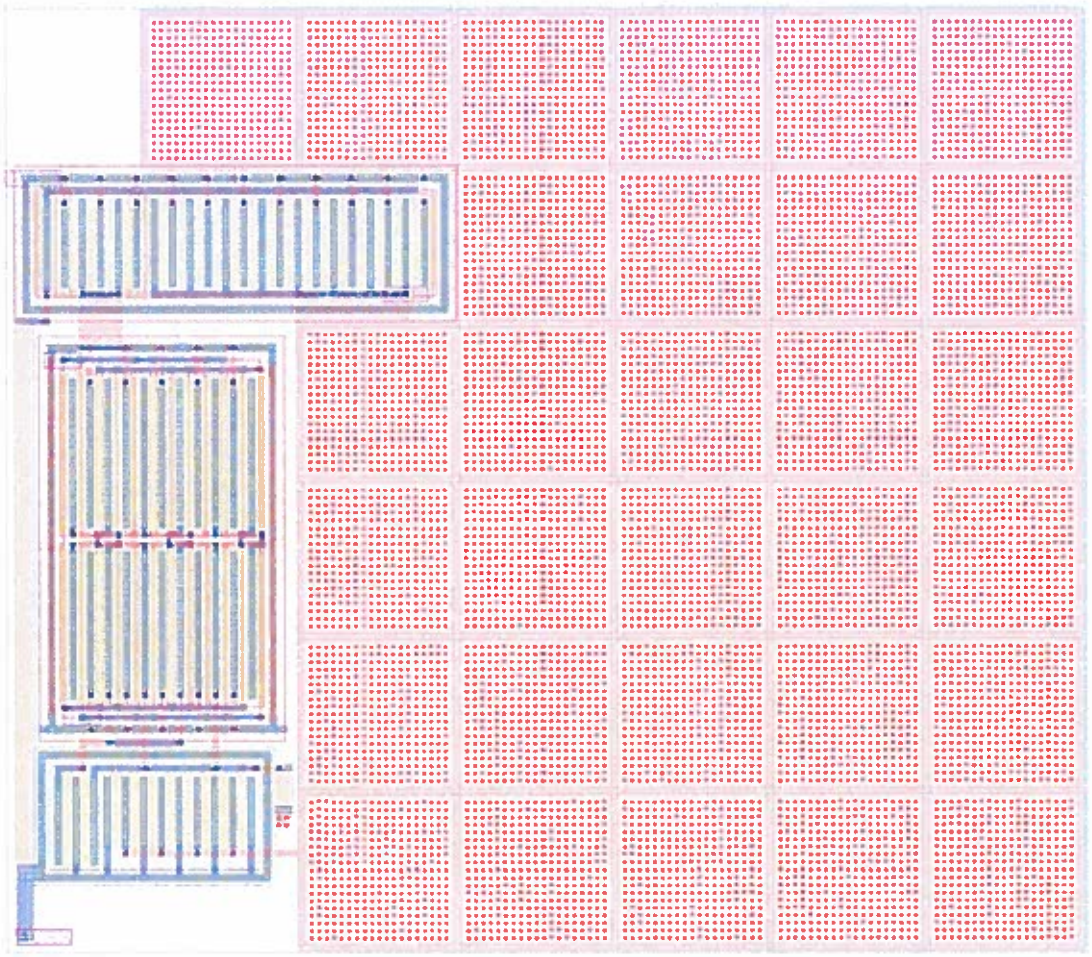
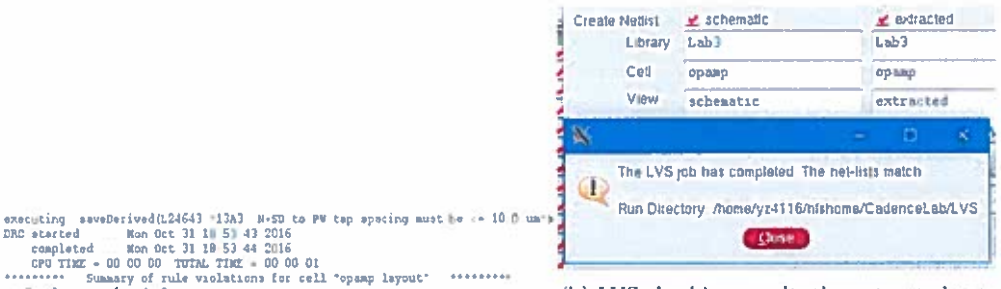


Figure 6: Layout design of the OTA



(a) DRC checking result, no errors

(b) LVS checking result, the extracted net-list matches the schematic

Figure 7: Layout design verification

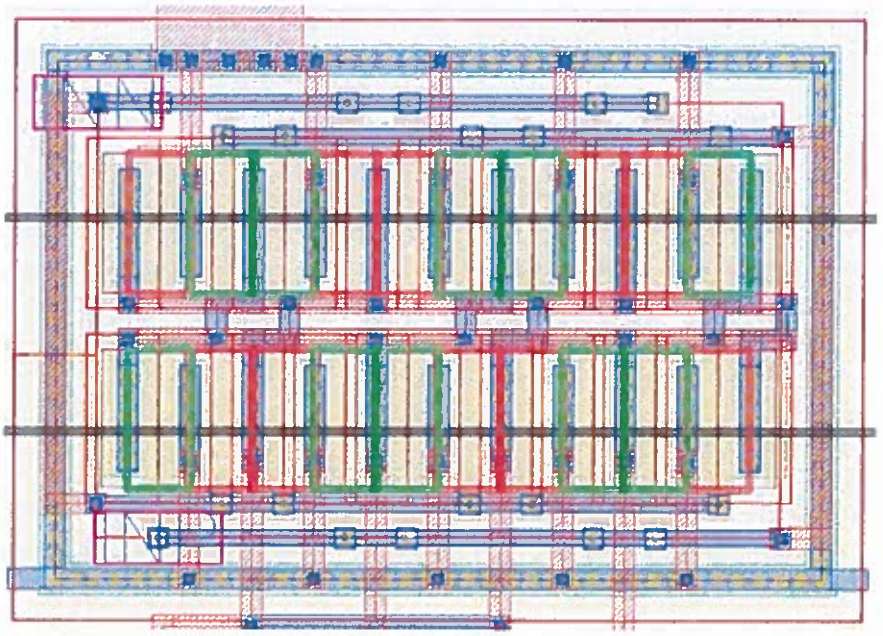


Figure 8: Layout of the LTP, vertically shrunk along the black lines

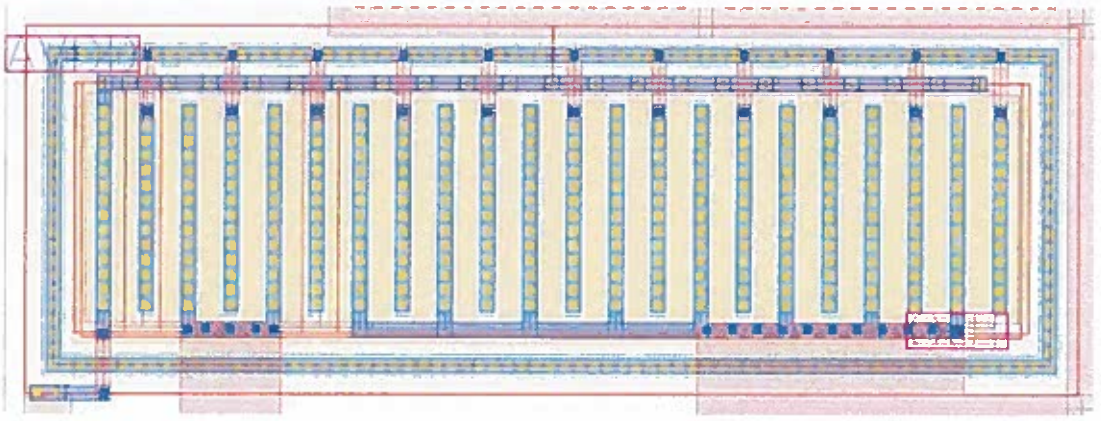


Figure 9: Layout of the PMOS current mirror

Lab 3 logbook submission_yz4116_attempt_2016-11-04-14-01-14
Name: Norman Zhi (yz4116)
Assignment: Lab 3 logbook submission
Date Submitted: Friday, 4 November 2016 14:01:14 o'clock GMT
Current Grade: 76.0

Submission Field:
There is no student submission text data for this assignment.

Comments:
There are no student comments for this assignment.

Files:
Original filename: logbook.pdf
Filename: Lab 3 logbook submission_yz4116_attempt_2016-11-04-14-01-14_logbook.pdf